

Through the loop of the program, it first returns key-value pairs of the precreated dictionary. Using the “importlib” library in the try-except block allows us to locate the missing packages, as “importlib.import_module(import_name)” will raise an error if not installed. The “except” will catch the error and append the package to a list of missing packages. Using assert, if the list of missing packages is not empty, a helpful message with the missing packages would be printed out. A message will also notify if all required packages are installed.

Below is the output of my program when not all packages are installed:

```
(my_env) brandonchau@MacBookPro hw0 % python python_basics_setup.py
Traceback (most recent call last):
  File "/Users/brandonchau/Desktop/School/Spring 2026/CECS 457/hw0/python_basics_setup.py", line 58, in <module>
    main()
  File "/Users/brandonchau/Desktop/School/Spring 2026/CECS 457/hw0/python_basics_setup.py", line 53, in main
    assert_required_packages()
  File "/Users/brandonchau/Desktop/School/Spring 2026/CECS 457/hw0/python_basics_setup.py", line 45, in assert_required_packages
    assert not missing_packages, (
           ~~~~~
AssertionError: Missing packages: matplotlib
Please install the list of packages before continuing.
```

Below is the output when all packages are installed:

```
(my_env) brandonchau@MacBookPro hw0 % python python_basics_setup.py
All required packages are installed and importable.
(my_env) brandonchau@MacBookPro hw0 %
```